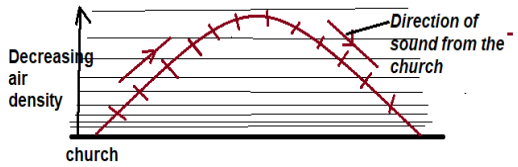
S.4 2026 PHYSICS

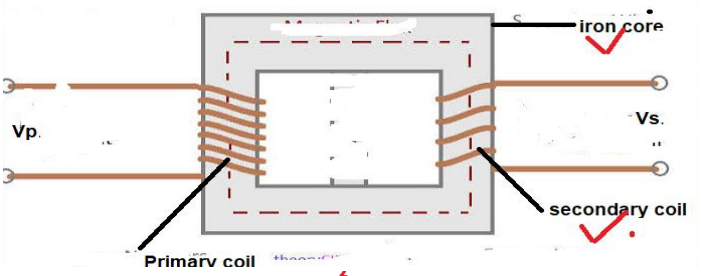
BUGANDA MOCKS

SCORE GUIDE

X/60

	Expected responses	Comment
ITEM ONE		
(a)	At night, *air near the ground is denser than that above*. Sound waves from the church travelling upwards undergoes *total internal reflection* and comes back to the ground making it seem louder  03	
(b)	The dress appears *red.* In red light, *the red colour is reflected* because it is common and no colour is absorbed In yellow light, which *contains red and green* , *Green colour is absorbed* and *red is reflected* . 05	
(c)	$c = f\lambda$ Red: $3.0 \times 10^8 = \mathrm{f}(7.5 \times 10^{-7}) \Leftrightarrow f = 4.0 \times 10^{14} m$ yellow: $3.0 \times 10^8 = f(5.8 \times 10^{-7}) \Leftrightarrow \mathrm{f} = 5.2 \times 10^{14} m$ *red colour should be used since it has a small wavelength* 04	
	Total	12
ITEM TWO		
(a)	$(450 - 50) \longrightarrow 200 \longrightarrow 100 \longrightarrow 50 \longrightarrow 25 \longrightarrow 12.5 \longrightarrow 6.25$ $6t = T \Longrightarrow 6(2) = 12 days$ The patient will remain in the hospital for 12 days 04	
(b)	Dangers of radioactive substances • Mutation • Eye irritation • Skin burn • Cancer • Infertility 02	Any two 1 score each
(c)	Handling radioactive substance • Use a pair of long tongs when carrying • Keep in a thick lead container • Wear protective gears • Keep away from if not in use • Don't eat/drink near it 02	Any two 1 score each
	Total	08

ITEM THREE		
	Moon appearance • The moon *reflects light from the sun*. • It *rotates/orbits the earth* and the amount of light it reflects depend on its phase. • These phases are; *new moon, waxing crescent, first quarter, waxing gibbous, full moon, wanning gibbous, third quarter and wanning crescent*. 03	
	Live TV airing • It is possible due to *(communication) satellite* • Videos and sound were encoded(converted into electric signals) and sent to the ground station • They were then sent to the *(geosynchronous) satellite* through *uplink channel* and *amplified* • They were then sent back the ground station near the receiver through downlink channel • The signals were the sent to the TV decoder for *decoding* back into video and sound 05	
	Cause of day and night lit is caused by *rotation of the earth* on its axis. The hemisphere of the earth *facing the sun experiences day time* The hemisphere of the earth *facing away from the sun experiences night time* 03	
	Why LED lights are preferred. • Affordable • Saves energy • Portable • Release less heat • Durable 02	Any two 1 score each
	Total	13
ITEM FOUR		
(a)	*Friction* force between the braking parts led to *heat generation* making the system feel hot. 02	
(b)	• The *bags loaded on the roof* *increased the C.O.G* making the car unstable on the road • Reduced by loading the bags in the *boot/inside the car* 03	
(c)	$v^2 = u^2 + 2as \Leftrightarrow 0^2 = 30^2 + 2a(30)$ $a = -15ms^{-2}$ $v = u + at \Leftrightarrow 0 = 30 + (-15)t$ $t = 2s$ The car tool 2 s to come to rest 04	
(d)	$Q = mc\Delta\theta = (30)(1200)(5) = 180{,}000J$	

	The energy absorbed by the system is 180,000J No heat is lost the surrounding 03	
	Total	12
ITEM FIVE		
(a)	$P = \frac{mgh}{t} \Rightarrow \frac{(1000)(0.200)(10)(8)}{60} = 266.7W$ The pump works at a rate of $266.7W$ 03	
	$Eff = \frac{work\ output}{work\ input} \times 100\%$ $Eff = \frac{266.7}{1800} \times 100\% = 15\% = 0.15$ The pump is not efficient since the ratio of useful work to energy input is less than 0.75 03	
	Due to *high SHC of water* which makes it to *heat up slowly during day time and slowly releases it at night.* By morning, *water still has some heat it absorbed the previous* day making it feel warm 03	
	Heat that evaporates water; $Q = ml_v = \left(\frac{600}{1000}\right)(2.56 \times 10^6) = 1{,}536{,}000J$ Heat that raise temp of evaporated water; $Q = mc\Delta\theta = \left(\frac{600}{1000}\right)(4200)(100 - 32) = 171{,}360J$ $1{,}536{,}000J$ evaporates water and $171{,}360J$ raises temp of evaporated water 03	
	Total	12
ITEM SIX		
	The machine is a *step down transformer* $\frac{V_p}{V_s} = \frac{N_p}{N_s} \Leftrightarrow \frac{240}{150} = \frac{120}{N_s} \Leftrightarrow N_s = 75$ The other coil has 75 turns 04	
	 *Ac enters the primary coil* This creates *changing magnetic field in the primary coil* The changing magnetic *field links up to the secondary coil*. A *lower voltage Vs is created in the secondary coil*. 06	

$$P = 960W = 0.960kW$$
$$Time = 4 \times 5 = 20hrs$$
$$Energy = 0.960 \times 20 = 19.2kWh$$
$$Cost = 19.2 \times 1000 = 19200/-$$

The budget is not sufficient since its more than 10,000/-

04

Caring and maintaining a rechargeable battery.

- Refilling of acid with water to maintain the acid level
- Regular charging
- Regular terminal cleaning
- Avoid over charge/discharge
- Avoid short circuit

01

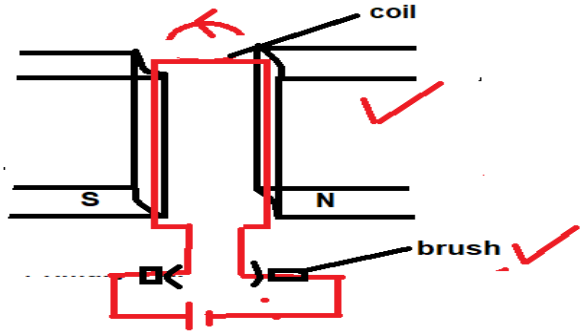
Any 1
1 score each

	Total	15

ITEM SEVEN

An electric drill contains an *electric motor*

Structure



Mode of operation

- Current is fed into the coil *creating magnetic field around it*
- These magnetic fields are *opposed* by those in the magnets
- This causes *the coil to rotate* hence spinning the drill

06

$$P = \frac{V^2}{R} \Rightarrow R = \frac{240^2}{12} = 96\Omega$$

The drill has a resistance of 96Ω

03

factor	Solution
Friction between moving parts generating heat and causing wear	lubrication
Resistance in the coils leading to heat generation	Use thick, shorter copper wires
Core loss/eddy currents leading to heat formation	Use iron core out of thin insulated iron layers
Air resistance leading to drag	Use aerodynamic roters

Any two plus solution

	04	
	Prevention of burn out • Using recommended voltage • Avoid continuous operation • Avoid hard surface 02	Any two
Total		15